Latihan 2.8

Tentukan deret fourier fungsi berikut !

1. $f(t) = \begin{cases} 0 & -5 < t < 0 \\ 1 & 0 < t < 5 \end{cases}$, dengan $f(t) = f(t + 10)$.

↳ $a_0 = \frac{1}{T} \int_0^T f(t) \cdot dt$

$= \frac{1}{10} \left(\int_{-5}^{0} (0) \cdot dt + \int_0^5 (1) \cdot dt \right)$

$= \frac{1}{10} \left(t \Big|_0^5 \right)$

$= \frac{1}{10} (5)$

$= \frac{5}{10} = \frac{1}{2}$

$w = \frac{2\pi}{T} = \frac{2\pi}{10} = \frac{\pi}{5}$

$a_n = \frac{2}{T} \int_0^{10} f(t) \cos nwt \cdot dt$

$= \frac{2}{10} \left(\int_{-5}^{0} (0) \cos \left(\frac{n\pi t}{5} \right) \cdot dt + \int_0^5 (1) \cos \left(\frac{n\pi t}{5} \right) \cdot dt \right)$

$= \frac{1}{5} \left(\frac{5 \sin \left(\frac{n\pi t}{5} \right)}{n\pi} \Big|_0^5 \right)$

$= \frac{1}{5} \left(\frac{5 \sin \left(\frac{n\pi 5}{5} \right)}{n\pi} \right)$

$= \frac{\sin (n\pi)}{n\pi}$

jika $n = 1 \rightarrow a_1 = 0$

$n = 2 \rightarrow a_2 = 0$ $\quad \therefore a_n = 0$

$n = 3 \rightarrow a_3 = 0$

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$$b_n = \frac{2}{T}\int_0^T f(t)\sin n\omega t \cdot dt$$

$$= \frac{2}{10}\left(\int_{-5}^{0} 0\cdot\sin\left(\frac{n\pi t}{5}\right)dt + \int_0^5 1\cdot\sin\left(\frac{n\pi t}{5}\right)dt\right)$$

$$= \frac{1}{5}\left(\frac{-\cos\left(\frac{n\pi t}{5}\right)}{n\pi}\Bigg|_0^5\right)$$

$$= \frac{1}{5}\left(\frac{-5\cos\left(\frac{n\pi 5}{5}\right)}{n\pi} + \frac{5\cos\left(\frac{n\pi(0)}{5}\right)}{n\pi}\right)$$

$$= \frac{1}{5}\left(-\frac{5\cos(n\pi)}{n\pi} + \frac{5}{n\pi}\right)$$

$$= -\frac{\cos(n\pi)}{n\pi} + \frac{1}{n\pi}$$

$$= \frac{1}{n\pi}\left(-\cos(n\pi)+1\right)$$

jika $n=1 \rightarrow b_1 = \frac{1}{\pi}\left(-\cos(\pi)+1\right) = \frac{2}{\pi}$

$n=2 \rightarrow b_2 = 0$

$n=3 \rightarrow b_3 = \frac{2}{3\pi}$

$n=4 \rightarrow b_4 = 0$

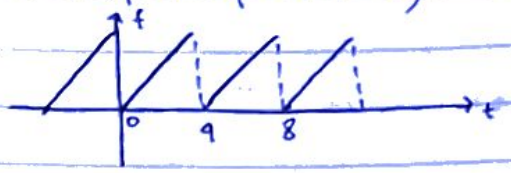
$\therefore b_n = \begin{cases} 0 \text{ ; genap} \\ \frac{2}{n\pi} \text{ ; ganjil} \end{cases}$

Jadi, $f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n\cos\frac{n\pi t}{5} + b_n\sin\frac{n\pi t}{5}\right)$ adalah

$$= \frac{1}{2} + \frac{2}{\pi}\sin t + \frac{2}{3\pi}\sin 3t + \frac{2}{5\pi}\sin 5t + \ldots$$

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10. Diketahui $f(t)$ seperti ditunjukkan pada gambar berikut !



$$f(t) = \begin{cases} T & 0 < t < 4 \\ 0 & 4 < t < 0 \end{cases} \qquad T = 4, \quad \omega = \frac{2\pi}{2} = \frac{\pi}{2}$$

$$a_0 = \frac{1}{T}\int_0^T f(t)\,dt$$

$$= \frac{1}{4}\int_0^4 \frac{t}{2}\,dt = \frac{1}{8}\left[\left(\frac{t^2}{2}\right)\Big|_0^4\right] = \frac{1}{8}(8-0) = 1$$

$$a_n = \frac{2}{4}\int_0^4 \frac{t}{2}\cos\frac{n\pi t}{2}\,dt$$

$$= \frac{1}{4}\int_0^4 t\cos\frac{n\pi t}{2}\,dt$$

$$= \frac{1}{4}\int_0^4 t\cos At\cdot dt \cdot \frac{1}{4}\left(\frac{\cos At}{A^2} + t\frac{\sin At}{A}\right)\Big|_0^4$$

$$= \frac{1}{4}\left(\frac{\cos\frac{n\pi}{2}t}{\left(\frac{n\pi}{2}\right)^2} + \frac{t\sin\frac{n\pi}{2}t}{\frac{n\pi}{2}}\right)\Big|_0^4$$

$$= \frac{1}{4}\left[\left(\frac{\cos 2n\pi}{\left(\frac{n\pi}{2}\right)^2} + \frac{4\sin 2n\pi}{\frac{n\pi}{2}}\right) - \left(\frac{1}{\left(\frac{n\pi}{2}\right)^2} + 0\right)\right]$$

$\therefore \cos 2n\pi = 1, \quad \sin 2n\pi = 0$

$$a_n = \frac{1}{4}\left(\frac{1}{\left(\frac{n\pi}{2}\right)^2} + 0 - \frac{1}{\left(\frac{n\pi}{2}\right)^2}\right) = 0$$

$$b_n = \frac{2}{4}\int_0^4 \frac{t}{2}\sin\frac{n\pi}{2}t\,dt$$

$$= \frac{2}{4}\cdot\frac{1}{2}\int_0^4 t\sin\frac{n\pi}{2}t\cdot dt$$

$$= \frac{1}{4}\int_0^4 \sin Bt\cdot dt$$

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$$= \frac{1}{4}\left(\frac{\sin Bt}{B^2} - \frac{t\cos Bt}{B}\right)\Bigg|_0^4$$

$$= \frac{1}{4}\left(\frac{\sin \frac{n\pi}{2}t}{\left(\frac{n\pi}{2}\right)^2} - \frac{t\cos \frac{n\pi}{2}t}{\frac{n\pi}{2}}\right)\Bigg|_0^4$$

$$= \frac{1}{4}\left[\left(\frac{\sin 2n\pi}{\left(\frac{n\pi}{2}\right)^2} - \frac{4\cos 2n\pi}{\frac{n\pi}{2}}\right) - (0-0)\right]$$

$$= \frac{1}{4}\left(0 - \frac{4}{\frac{n\pi}{2}}\right)$$

$$= \frac{-8}{4n\pi}$$

$$= -\frac{2}{n\pi}$$

$b_n = -\frac{2}{n\pi}$ (untuk n ganjil dan genap)

Jadi, $f(t) = 1 + \left(-\frac{2}{\pi}\sin\frac{\pi}{2}t - \frac{2}{2\pi}\sin\frac{2\pi}{2}t - \frac{2}{3\pi}\sin\frac{3\pi}{2}t - \frac{2}{4}\sin\frac{4\pi}{2}t\right)$

$$= 1 - \frac{2}{\pi}\left(\sin\frac{\pi}{2}t + \frac{1}{2}\sin\pi t + \frac{1}{3}\sin\frac{3\pi}{2}t + \frac{1}{4}\sin 2\pi t\right)$$